

HBase and Hive Lab Detailed Solutions

Section A: HBase Programming

1. HBase Table Operations:

Step 1: Open HBase shell using the command:

```
hbase shell
```

Step 2: Create a table named 'Employee':

```
create 'Employee', 'PersonalDetails', 'ProfessionalDetails'
```

Step 3: Insert at least 5 rows of data:

```
put 'Employee', '1', 'PersonalDetails:name', 'John'
put 'Employee', '1', 'ProfessionalDetails:designation', 'Manager'
put 'Employee', '2', 'PersonalDetails:name', 'Alice'
put 'Employee', '2', 'ProfessionalDetails:designation', 'Developer'
put 'Employee', '3', 'PersonalDetails:name', 'Bob'
put 'Employee', '3', 'ProfessionalDetails:designation', 'Manager'
put 'Employee', '4', 'PersonalDetails:name', 'Eve'
put 'Employee', '4', 'ProfessionalDetails:designation', 'Analyst'
put 'Employee', '5', 'PersonalDetails:name', 'Charlie'
put 'Employee', '5', 'ProfessionalDetails:designation', 'HR'
```

Step 4: Retrieve and display records:

```
scan 'Employee'
```

2. HBase Program with Filters

Step 1: Open HBase shell:

```
hbase shell
```

Step 2: Use the same table created previously ('Employee').

Step 3: Use SingleColumnValueFilter to display only rows where designation is 'Manager':

```
scan 'Employee', {
```

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```
FILTER => "SingleColumnValueFilter('ProfessionalDetails', 'designation', =, 'binary:Manager')"  
}
```

Section B: Hive Programming

3. Basic Table Operations in Hive:

Step 1: Open Hive shell:

```
hive
```

Step 2: Create database and switch to it:

```
CREATE DATABASE companyDB;  
USE companyDB;
```

Step 3: Create table:

```
CREATE TABLE company (  
    cmp_id INT,  
    cmp_name STRING,  
    cmp_palice FLOAT,  
    date_started STRING,  
    awards INT  
);
```

Step 4: Insert records:

```
INSERT INTO company VALUES (1, 'TCS', 25000, '2010-04-01', 5);  
INSERT INTO company VALUES (2, 'Infosys', 27000, '2011-05-20', 3);  
INSERT INTO company VALUES (3, 'Wipro', 22000, '2008-03-12', 4);  
INSERT INTO company VALUES (4, 'IBM', 30000, '2009-06-30', 2);  
INSERT INTO company VALUES (5, 'Accenture', 26000, '2012-07-14', 6);
```

Step 5: Display all records:

```
SELECT * FROM company;
```

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4. Hive Table with HDFS Upload

Step 1: Create a database and switch to it:

```
CREATE DATABASE companyDB;
```

```
USE companyDB;
```

Step 2: Create table:

```
CREATE TABLE employee (  
    emp_id INT,  
    emp_name STRING  
);
```

Step 3: Prepare a text file like:

101 John

102 Alice

103 Bob

104 Eve

105 Charlie

Step 4: Upload file to HDFS:

```
hdfs dfs -put employee.txt /user/hive/warehouse/
```

Step 5: Create Hive table for TEXTFILE:

```
CREATE TABLE employee_hdfs (  
    emp_id INT,  
    emp_name STRING  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY '\t'  
STORED AS TEXTFILE;
```

Step 6: Load data into Hive:

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```
LOAD DATA INPATH '/user/hive/warehouse/employee.txt' INTO TABLE employee_hdfs;
```

Step 7: Verify:

```
SELECT * FROM employee_hdfs;
```

5. Complex Hive Query

Step 1: Create database and use it:

```
CREATE DATABASE companyDB;  
USE companyDB;
```

Step 2: Create MARKETING table:

```
CREATE TABLE MARKETING (  
    emp_id INT,  
    emp_name STRING,  
    designation STRING,  
    salary FLOAT,  
    profit FLOAT  
);
```

Step 3: Insert data into MARKETING:

```
INSERT INTO MARKETING VALUES  
(1, 'John', 'Manager', 90000, 60000),  
(2, 'Alice', 'Developer', 75000, 40000),  
(3, 'Bob', 'Manager', 85000, 55000),  
(4, 'Eve', 'Analyst', 70000, 30000),  
(5, 'Charlie', 'HR', 65000, 45000);
```

Step 4: Display employees with profit > 50000:

```
SELECT * FROM MARKETING WHERE profit > 50000;
```

Step 5: Group by designation and average salary:

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```
SELECT designation, AVG(salary) FROM MARKETING GROUP BY designation;
```

Step 6: Retrieve top 3 highest paid employees:

```
SELECT * FROM MARKETING ORDER BY salary DESC LIMIT 3;
```

Step 7 (Bonus): Join with department table:

```
CREATE TABLE department (  
    dept_id INT,  
    dept_name STRING  
);
```

```
INSERT INTO department VALUES (1, 'HR'), (2, 'IT'), (3, 'Marketing');
```

```
SELECT M.emp_name, D.dept_name  
FROM MARKETING M  
JOIN department D ON M.emp_id = D.dept_id;
```